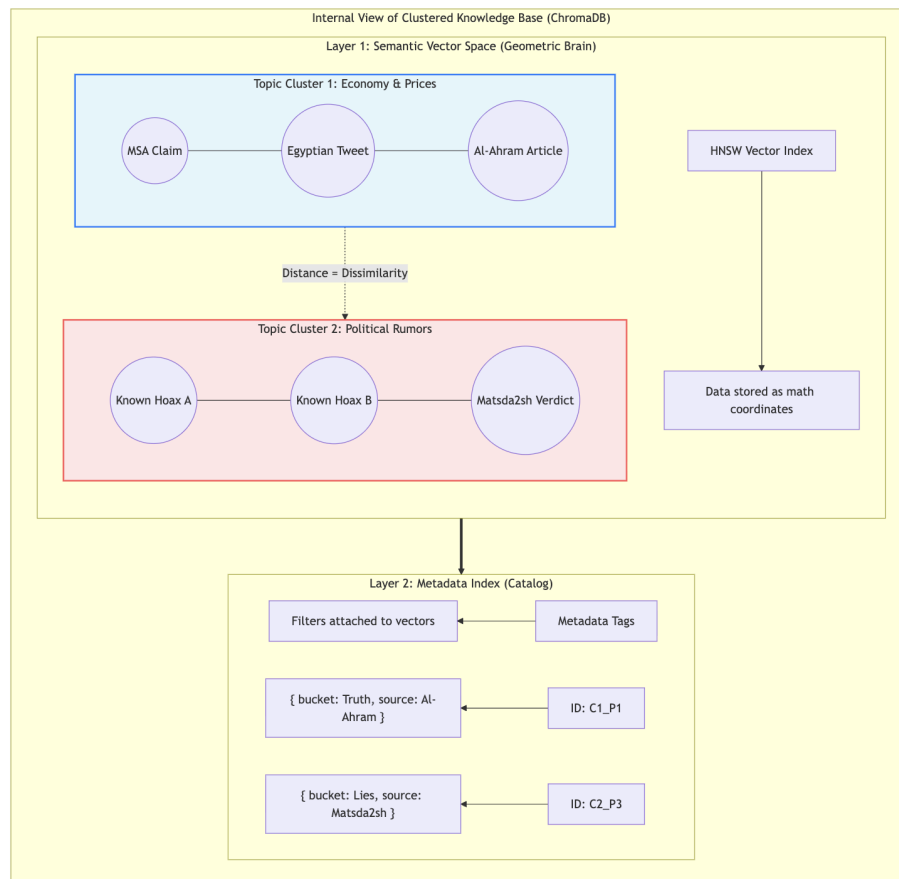


System 2: Automated Fact-Verification & Knowledge Retrieval

1. Architectural Overview

System 2 is an analytical engine designed to verify claims by cross-referencing social media input with a dual-bucket Knowledge Base (KB). It bridges the gap between **Egyptian Dialect** and **Modern Standard Arabic (MSA)** evidence.



2. The Dual-Bucket Knowledge Base

- **Bucket A (Verified Claims):** Contains previously debunked rumors and fact-checks. Stores both Dialect and MSA versions to allow for "Direct Slang Matching."
- **Bucket B (Global Truth):** A live-ingested repository of 1,000+ high-authority news articles (RT, Youm7, Al-Ahram), surgically cleaned of UI noise.

3. Hierarchical Multi-Step Retrieval Logic

To optimize performance and accuracy, the system follows a three-stage "Smart Search" protocol:

1. **Stage 1: Priority Slang Search (Bucket A)** * The system first attempts a semantic match in the original dialect. If a "Direct Hit" is found in Bucket A, the verdict is returned instantly, bypassing unnecessary translation.
2. **Stage 2: Dialect Mapping & Deep Search (Bucket B)**
 - If no match exists, the claim is mapped to its MSA semantic equivalent (using MARBERT/AraBERT) to query the formal news articles in Bucket B.
3. **Stage 3: Recursive Evidence Digging**
 - If evidence is found but confidence is low, the system extracts entities and performs a multi-step search to find secondary supporting sources.

4. KB Structure & Semantic Clustering

As illustrated in our technical design, data is organized into a **Semantic Vector Space (ChromaDB)**.

- **Clustering:** Claims and articles are grouped by topic proximity (e.g., Economy, Health).
- **Mapping:** The system ensures that a dialectal claim and its MSA news counterpart map to the same geometric point.

5. Mini Proof of Work (Implementation Status)

A. Knowledge Base Milestone

- **Status:** Completed.
- **Metric:** 1,067 unique records discovered; 769 "Gold Standard" surgically cleaned articles successfully ingested into Bucket B.